Polynomials over the integers

L5-AL1 Level 5 Advanced · Algebra

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What integer coefficients force: divisibility of differences and periodic points, the binomial basis, Gauss, Eisenstein and root-size irreducibility, roots of unity, and Schur's theorem on prime divisors.
By the end you can use a - b | P(a) - P(b), difference tables and the binomial basis to decide which values an integer polynomial can take; prove irreducibility with Gauss's lemma, Eisenstein (after a shift) and estimates on the size of roots; use cyclotomic polynomials and Schur's theorem, combined with the Chinese remainder theorem, in olympiad proofs.
Prerequisites: Polynomials and Vieta's formulas (L3-AL1), Orders, Fermat and Euler (L4-NT1).
Contests: BMO Round 2, EGMO and IMO selection, USAMO, IMO. **Time:** about 180 minutes.

1 Lesson

Is there a polynomial P with integer coefficients such that $P(2) = 5$ and $P(9) = 17$? No. For every $k \geq 1$ the number $9^k - 2^k$ is divisible by $9 - 2 = 7$, so $P(9) - P(2)$ is a multiple of 7, while $17 - 5 = 12$ is not. This one-line argument is the first of several facts in this unit that hold for polynomials with integer coefficients and fail for polynomials with rational or real coefficients. We assume L3-AL1 (Vieta, the factor theorem) and the modular arithmetic of L4-NT1 throughout. Write $\mathbb{Z}[x]$ for the set of polynomials with integer coefficients.

1.1 Differences of values

Theorem 1.
If $P \in \mathbb{Z}[x]$ and a, b are integers, then $a - b \mid P(a) - P(b)$. In particular, $P(n) \bmod m$ depends only on $n \bmod m$.
Proof. $P(a) - P(b) = \sum_k c_k(a^k - b^k)$ and $a^k - b^k = (a-b)(a^{k-1} + a^{k-2}b + \cdots + b^{k-1})$.

Two further facts about $\mathbb{Z}[x]$ are used constantly. If $D \in \mathbb{Z}[x]$ is *monic*, long division of $P \in \mathbb{Z}[x]$ by D never divides by anything except 1, so the quotient and remainder lie in $\mathbb{Z}[x]$. Hence if $P \in \mathbb{Z}[x]$ vanishes at the integers $r_1, \ldots, r_k$, then $P(x) = (x - r_1) \cdots (x - r_k)Q(x)$ with $Q \in \mathbb{Z}[x]$.

Example 1.
A polynomial P with integer coefficients satisfies $P(P(P(5))) = 5$. Prove that $P(5) = 5$.

Solution.
We know nothing about P except one composite value, so the only usable fact is Theorem 1, applied along the orbit. Put $x_0 = 5$ and $x_{i+1} = P(x_i)$, so $x_3 = x_0$, and let $d_i = x_{i+1} - x_i$. Note that $x_4 = P(x_3) = P(x_0) = x_1$, so $d_3 = d_0$.
By Theorem 1, $d_i = x_{i+1} - x_i$ divides $P(x_{i+1}) - P(x_i) = d_{i+1}$. Hence $d_0 \mid d_1 \mid d_2 \mid d_3 = d_0$, and so $|d_0| = |d_1| = |d_2| = d$, say. But $d_0 + d_1 + d_2 = x_3 - x_0 = 0$, and a sum of three numbers each equal to $\pm d$ is an odd multiple of d. Hence $d = 0$, that is, $P(5) = x_1 = x_0 = 5$.

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The same argument with k steps proves the general statement, which is worth remembering.

Theorem 2 (periodic points).
Let $P \in \mathbb{Z}[x]$, $x_0 \in \mathbb{Z}$ and $x_{i+1} = P(x_i)$. If $x_k = x_0$ for some $k \geq 1$, then $x_2 = x_0$.
Proof. With $d_i = x_{i+1} - x_i$ as above, the sequence (d_i) has period k and $d_i \mid d_{i+1}$. If some $d_i = 0$ then all later d_j are 0, so by periodicity all $d_i = 0$ and $x_1 = x_0$. Otherwise $|d_0| \leq |d_1| \leq \cdots \leq |d_k| = |d_0|$, so all $|d_i|$ are equal. Since $d_0 + \cdots + d_{k-1} = 0$, the signs are not all equal, so $d_{i+1} = -d_i$ for some i, which means $x_{i+2} = x_i$. Then $x_{j+2} = x_j$ for all $j \geq i$, and by periodicity for all j, in particular $j = 0$.

1.2 Integer-valued polynomials and the binomial basis

Theorem 1 is necessary for integer coefficients but not sufficient, and the right tool for deciding exactly which values a polynomial in $\mathbb{Z}[x]$ can take is the basis $\binom{x}{k} = \frac{x(x-1)\cdots(x-k+1)}{k!}$. Write $\Delta P(x) = P(x+1) - P(x)$. Pascal's rule gives $\Delta\binom{x}{k} = \binom{x}{k-1}$.

Theorem 3.
Every polynomial P of degree at most n (with rational, real or complex coefficients) can be written uniquely as $P(x) = \sum_{k=0}^{n} c_k \binom{x}{k}$, and $c_k = \Delta^k P(0)$. Moreover:

1. $P(m) \in \mathbb{Z}$ for every integer m if and only if every c_k is an integer, if and only if $P(0), P(1), \ldots, P(n)$ are integers;
2. $P \in \mathbb{Z}[x]$ if and only if $k! \mid c_k$ for every k.

Proof. Since $\binom{x}{k}$ has degree k, these polynomials form a basis, and applying Δ^j and setting $x = 0$ kills every term except c_j. For (1): if $P(0), \ldots, P(n)$ are integers, so are all differences $\Delta^j P(0)$; conversely $\binom{m}{k}$ is an integer for every integer m (for $m < 0$ note $\binom{m}{k} = (-1)^k\binom{k-m-1}{k}$). For (2): if $k! \mid c_k$ then $c_k\binom{x}{k} = \frac{c_k}{k!}x(x-1)\cdots(x-k+1) \in \mathbb{Z}[x]$. Conversely, if $P \in \mathbb{Z}[x]$ has degree n and leading coefficient a, comparing leading coefficients gives $c_n = n!\,a$, and $P - a\,x(x-1)\cdots(x-n+1)$ lies in $\mathbb{Z}[x]$ and has smaller degree; induct.

Note that for a polynomial in $\mathbb{Z}[x]$ of any degree, $c_k = \Delta^k P(0)$ for $k \leq n$ depends only on $P(0), \ldots, P(n)$. Hence values $v_0, \ldots, v_n$ are taken at $0, \ldots, n$ by some $P \in \mathbb{Z}[x]$ if and only if $k! \mid \Delta^k v(0)$ for $k = 0, \ldots, n$.

Example 2.
Find all integers t for which some polynomial with integer coefficients satisfies $P(0) = 3$, $P(1) = 1$, $P(2) = 3$, $P(3) = 15$ and $P(4) = t$.

Solution.
First try Theorem 1 on each pair: $4 \mid t-3$, $3 \mid t-1$, $2 \mid t-3$ and the rest hold automatically, so $t \equiv 7 \pmod{12}$. This is necessary, but we do not yet know it is sufficient. The difference table is:

$$\begin{array}{ccccccccc}
3 & & 1 & & 3 & & 15 & & t \\
& -2 & & 2 & & 12 & & t-15 & \\
& & 4 & & 10 & & t-27 & & \\
& & & 6 & & t-37 & & & \\
& & & & t-43 & & & &
\end{array}$$

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So $c_0 = 3$, $c_1 = -2$, $c_2 = 4$, $c_3 = 6$, $c_4 = t - 43$. The conditions $1 \mid c_1$, $2 \mid c_2$, $6 \mid c_3$ hold, and $24 \mid c_4$ is needed. Hence $t \equiv 43 \equiv 19 \pmod{24}$. Conversely, for such t the polynomial $3 - 2\binom{x}{1} + 4\binom{x}{2} + 6\binom{x}{3} + (t-43)\binom{x}{4}$ lies in $\mathbb{Z}[x]$ by Theorem 3 and takes the required values. The answer is $t \equiv 19 \pmod{24}$; in particular $t = 7$ passes every test from Theorem 1 and still fails.

1.3 Irreducibility

A polynomial in $\mathbb{Z}[x]$ is *primitive* if the highest common factor of its coefficients is 1.

Theorem 4 (Gauss's lemma).
The product of two primitive polynomials is primitive. Hence if $f \in \mathbb{Z}[x]$ factors as a product of two nonconstant polynomials in $\mathbb{Q}[x]$, it factors as a product of two nonconstant polynomials in $\mathbb{Z}[x]$ of the same degrees; if f is monic, both factors can be taken monic.
Proof. Let $g = \sum b_i x^i$, $h = \sum c_j x^j$ be primitive and p a prime. Take the least i with $p \nmid b_i$ and the least j with $p \nmid c_j$. The coefficient of x^{i+j} in gh is $b_i c_j$ plus terms each containing some $b_{i'}$ with $i' < i$ or $c_{j'}$ with $j' < j$, so it is not divisible by p. For the second part, write a rational factorisation as $f = \frac{r}{s} g_0 h_0$ with g_0, h_0 primitive; then $sf = r g_0 h_0$, and comparing contents gives $\frac{r}{s} \in \mathbb{Z}$.

Theorem 5 (Eisenstein).
Let $f = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ and p a prime with $p \mid a_0, \ldots, a_{n-1}$, $p \nmid a_n$ and $p^2 \nmid a_0$. Then f is irreducible over $\mathbb{Q}$.
Proof. By Theorem 4 suppose $f = gh$ with $g = \sum_{i \le r} b_i x^i$, $h = \sum_{j \le s} c_j x^j$ in $\mathbb{Z}[x]$ and $r, s \ge 1$. Exactly one of b_0, c_0 is divisible by p, say $p \mid b_0$. Since $p \nmid a_n = b_r c_s$, $p \nmid b_r$, so there is a least $i \le r < n$ with $p \nmid b_i$. Then $a_i = b_i c_0 + b_{i-1} c_1 + \cdots \equiv b_i c_0 \not\equiv 0 \pmod{p}$, contradicting $p \mid a_i$.

Eisenstein often applies only after the substitution $x \mapsto x + 1$ (or $x \mapsto x + a$), which does not change irreducibility. When no prime works, look at the sizes of the complex roots instead.

Theorem 6 (small constant term).
Let $f \in \mathbb{Z}[x]$ be monic with $|f(0)|$ prime, and suppose every complex root z of f has $|z| > 1$. Then f is irreducible over $\mathbb{Q}$.
Proof. If $f = gh$ with g, h monic, nonconstant and in $\mathbb{Z}[x]$, then $|g(0)|$ is the product of the moduli of the roots of g, so $|g(0)| > 1$, and similarly $|h(0)| > 1$. These are integers, so $|f(0)| = |g(0)||h(0)|$ is a product of two integers greater than 1, which is not prime.

Example 3.
Determine all integers $n \ge 2$ for which $x^n + x + 2$ is irreducible over $\mathbb{Q}$.

Solution.
Eisenstein at 2 fails because the coefficient of x is 1. The constant term is the prime 2, so Theorem 6 is the thing to try: we need to know whether a root can satisfy $|z| \le 1$. If $|z| \le 1$ then $|z^n + z| \le |z|^n + |z| \le 2$, and $z^n + z = -2$ forces equality throughout: $|z| = 1$ and z^n, z point in the same direction, so $z^n = z$ and $2z = -2$, that is $z = -1$ and $(-1)^n = -1$.
Hence if n is even, every root has $|z| > 1$ and the polynomial is irreducible by Theorem 6. If

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n is odd, $(-1)^n - 1 + 2 = 0$, so $x+1$ is a factor and, as $n \geq 2$, the polynomial is reducible. The answer is: exactly the even n. Note that the equality case of the triangle inequality is where the parity of n enters; a solution that only says $|z^n + z| \leq 2$ proves nothing.

1.4 Roots of unity

The cyclotomic polynomials are defined by $x^n - 1 = \prod_{d|n} \Phi_d(x)$, so Φ_n is the monic polynomial whose roots are the primitive nth roots of unity. By induction on n and division by monic polynomials, $\Phi_n \in \mathbb{Z}[x]$. For a prime p, $\Phi_p(x) = 1 + x + \cdots + x^{p-1}$, and

$$\Phi_p(x+1) = \frac{(x+1)^p - 1}{x} = x^{p-1} + \binom{p}{p-1} x^{p-2} + \cdots + \binom{p}{1}$$

satisfies Eisenstein at p, so Φ_p is irreducible. (Every Φ_n is irreducible over $\mathbb{Q}$; this is a theorem of Gauss whose proof for composite n is longer, and I will not use it.)

Theorem 7.
Let p be prime, $\zeta = e^{2\pi i/p}$, and $a_0, \ldots, a_{p-1}$ rationals with $a_0 + a_1\zeta + \cdots + a_{p-1}\zeta^{p-1} = 0$. Then $a_0 = a_1 = \cdots = a_{p-1}$.
Proof. Let $A(x) = \sum a_i x^i$ and $D = \gcd(A, \Phi_p)$ in $\mathbb{Q}[x]$. Since $D = UA + V\Phi_p$ for some $U, V \in \mathbb{Q}[x]$, $D(\zeta) = 0$, so D is a nonconstant factor of the irreducible Φ_p. Hence $\Phi_p \mid A$, and as $\deg A \leq p-1$, $A = c\,\Phi_p$.

We also need the fundamental theorem of symmetric polynomials: a symmetric polynomial with integer coefficients in $z_1, \ldots, z_n$ is an integer polynomial in the elementary symmetric functions. So if $z_1, \ldots, z_n$ are the roots of a monic $P \in \mathbb{Z}[x]$, every such expression is an integer.

1.5 Prime divisors of values

Theorem 8 (Schur).
If $P \in \mathbb{Z}[x]$ is nonconstant, infinitely many primes divide at least one of the nonzero numbers among $P(1), P(2), P(3), \ldots$
Proof. If $P(0) = 0$, every prime p divides $P(p)$, and $P(p) \neq 0$ for all but finitely many p. Otherwise let $c = P(0) \neq 0$ and suppose only the primes $p_1, \ldots, p_r$ occur; put $N = p_1 \cdots p_r$. Every nonconstant term of $P(|c|Nx)$ is divisible by cNx, so $P(|c|Nx) = c\,(1 + Nx\,Q(x))$ with $Q \in \mathbb{Z}[x]$, and $1 + NxQ(x)$ is nonconstant. It equals ± 1 for only finitely many x, so choose a positive integer x with $|1 + NxQ(x)| > 1$ and a prime q dividing it. Then $P(|c|Nx) \neq 0$ and $q \mid P(|c|Nx)$, but $1 + NxQ(x) \equiv 1 \pmod{p_i}$ for every i, so q is a new prime. (We use $|c|$ rather than c so that the argument $|c|Nx$ is a positive integer.)

Example 4.
Let $P \in \mathbb{Z}[x]$ be nonconstant and k a positive integer. Prove that there is a positive integer n such that $P(n)$ has at least k distinct prime factors.

Solution.
Schur gives many primes, but each comes with its own argument a; we need one n for all of them at once, so we combine the congruences with the Chinese remainder theorem. By Theorem 8 choose distinct primes $p_1, \ldots, p_k$ and positive integers $a_1, \ldots, a_k$ with $p_i \mid P(a_i)$.

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By CRT there is n_0 with $n_0 \equiv a_i \pmod{p_i}$ for all i, and then every $n = n_0 + mp_1 \cdots p_k$ satisfies the same congruences. By Theorem 1, $P(n) \equiv P(a_i) \equiv 0 \pmod{p_i}$ for each i. Finally P has finitely many roots, so we may choose m with $n > 0$ and $P(n) \neq 0$, and then $P(n)$ has the k distinct prime factors $p_1, \ldots, p_k$.
The condition $P(n) \neq 0$ is the step most scripts omit: without it, "$p_i \mid P(n)$ for all i" says nothing.

Common traps.

- Treating $a - b \mid P(a) - P(b)$ as a complete test for which values are possible. It is not (Example 2); use differences and Theorem 3.
- Dividing by a non-monic polynomial and assuming the quotient has integer coefficients.
- Applying Eisenstein without checking the two end conditions: p must not divide the leading coefficient, and p^2 must not divide the constant term.
- Proving that f has no rational root and concluding that f is irreducible. This is valid only in degrees 2 and 3.
- In root-size arguments, using $\leq$ where the lemma needs $<$, and not analysing the equality case.

1.6 How to spot this in a contest

The phrase "polynomial with integer coefficients" is a signal to use Theorem 1 at once, on every pair of arguments in sight. My order of attack is:

1. Values at several integers, or iterates such as $P(P(a))$: differences of arguments divide differences of values; for orbits, Theorem 2.
2. Which values are possible, or "$P(n)$ is an integer for all n": the binomial basis and difference tables.
3. Irreducibility: Eisenstein, then Eisenstein after $x \mapsto x + a$, then root sizes against a prime constant term; factor as gh with g, h monic in $\mathbb{Z}[x]$ and compare $g(0), h(0)$, or $g(a), h(a)$ at a cleverly chosen a.
4. Sums of powers of a root of unity, or counting modulo a prime: Theorem 7 and the factorisation of $x^p - 1$.
5. "Infinitely many primes" or "$P(n)$ divides something for all n": Schur, then CRT to combine primes.

Problems at BMO2 and IMO level usually need two of these together, for example Schur followed by Theorem 1 in the form $P(n + p) \equiv P(n) \pmod{p}$.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

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Problem 1 ⋆⋆ Let P be a polynomial with integer coefficients and $a \neq b$ integers such that $P(a)P(b) = -(a-b)^2$. Prove that $P(a) + P(b) = 0$.

Problem 2 ⋆⋆ How many polynomials P with integer coefficients and degree at most 4 satisfy $P(0), P(1), P(2), P(3), P(4) \in \{0, 1, 2, 3\}$?

Problem 3 ⋆⋆⋆ Let p be an odd prime. Prove that

$$f(x) = x^{p-2} + 2x^{p-3} + 3x^{p-4} + \cdots + (p-2)x + (p-1)$$

is irreducible over $\mathbb{Q}$.

Problem 4 ⋆⋆⋆ Find all polynomials P with integer coefficients such that $P(n)$ divides n^n for every positive integer n.

Problem 5 ⋆⋆⋆⋆ Let P be a monic polynomial with integer coefficients, all of whose complex roots have absolute value 1. Prove that every root of P is a root of unity.

Problem 6 ⋆⋆⋆⋆ Let P be a nonconstant polynomial with integer coefficients and k a positive integer. Prove that there are k consecutive positive integers $n+1, n+2, \ldots, n+k$ such that each of $P(n+1), \ldots, P(n+k)$ is divisible by the square of some prime.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **USAMO 1974 Problem 1** ⋆⋆
   Differences of arguments divide differences of values; go round the cycle.
2. **IMO 1993 Problem 1** ⋆⋆⋆
   Gauss's lemma, then compare the constant terms of the factors and the moduli of their roots.
3. **USAMO 1976 Problem 5** ⋆⋆⋆
   Substitute the primitive fifth roots of unity.
4. **BMO2 2013/14 Problem 3** ⋆⋆⋆
   Each term is a value of x^2 - x at the previous term; look at coprime factors for (a) and at residues for (b).
5. **BMO2 2018/19 Problem 3** ⋆⋆⋆
   Encode subset sums in a product of factors $1 + x^k$ and evaluate at a pth root of unity.
6. **BMO2 2011/12 Problem 4** ⋆⋆⋆
   The power sums of the roots of an integer polynomial satisfy a linear recurrence with integer coefficients.
7. **USAMO 2012 Problem 4** ⋆⋆⋆⋆
   Use the divisibility condition with a large factorial against small values.
8. **USAMO 1995 Problem 4** ⋆⋆⋆⋆
   Interpolate the first few terms in the binomial basis and show the difference vanishes.
9. **IMO 2006 Problem 5** ⋆⋆⋆⋆
   Periodic points of an integer polynomial have period at most 2; then count solutions of two polynomial equations.

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4 Hints

1. Compare the power of each prime p in $P(a)$, $P(b)$ and $a-b$, using $a-b \mid P(a)-P(b)$.
2. Each choice of the five values gives exactly one polynomial of degree at most 4. Use Theorem 1 to cut down the choices, then test the fourth difference.
3. Compute $(x-1)^2 f(x)$, then substitute $x \mapsto x+1$.
4. Write $P(x) = x^k Q(x)$ with $Q(0) \neq 0$, and look at $Q(q)$ for a large prime q.
5. For each $k \geq 1$ consider the monic polynomial whose roots are the kth powers of the roots of P. How many such polynomials can there be?
6. Take an irreducible factor R of P; then R and R' have no common factor. Find primes p and integers b with $p^2 \mid R(b)$, then use CRT.

5 Solutions

Problem 1

Let P be a polynomial with integer coefficients and $a \neq b$ integers such that $P(a)P(b) = -(a-b)^2$. Prove that $P(a)+P(b) = 0$.

Let $u = P(a)$, $v = P(b)$ and $d = a-b \neq 0$. Then $uv = -d^2 \neq 0$, so $u, v \neq 0$, and by Theorem 1, $d \mid u-v$.

Fix a prime p and let α, β, δ be the exponents of p in u, v, d. From $uv = -d^2$ we get $\alpha+\beta = 2\delta$. If $\alpha \neq \beta$, then the exponent of p in $u-v$ is $\min(\alpha, \beta)$, which is less than $\frac{\alpha+\beta}{2} = \delta$; this contradicts $d \mid u-v$. Hence $\alpha = \beta = \delta$.

This holds for every prime, so $|u| = |v| = |d|$. Since $uv = -d^2 < 0$, u and v have opposite signs, and therefore $u = -v$, that is, $P(a)+P(b) = 0$.

Problem 2

How many polynomials P with integer coefficients and degree at most 4 satisfy $P(0), P(1), P(2), P(3), P(4) \in \{0,1,2,3\}$?

By Lagrange interpolation each choice of values $v_i = P(i)$ gives exactly one polynomial of degree at most 4, so we count the tuples $(v_0, \ldots, v_4)$ that come from a polynomial in $\mathbb{Z}[x]$. By Theorem 3 (and the remark after it) this happens if and only if $k! \mid \Delta^k v(0)$ for $k \leq 4$.

First we use Theorem 1. Since $4 \mid v_4 - v_0$, $v_4 = v_0 = a$, say. Since $3 \mid v_3 - v_0$ and $3 \mid v_4 - v_1$, both v_1 and v_3 are congruent to a modulo 3; since $2 \mid v_3 - v_1$, they have the same parity. If $a \in \{1,2\}$ this forces $v_1 = v_3 = a$; if $a \in \{0,3\}$ it forces $v_1 = v_3 = b$ with $b \in \{0,3\}$. Finally $v_2 \equiv a \pmod 2$. This leaves twelve tuples (a,b,c,b,a), with $b = a$ when $a \in \{1,2\}$.

The fourth difference is $\Delta^4 v(0) = v_4 - 4v_3 + 6v_2 - 4v_1 + v_0 = 2a - 8b + 6c$, which must be divisible by 24. For $a = b = 1$, $c \in \{1,3\}$ gives 0 or 12; for $a = b = 2$, $c \in \{0,2\}$ gives -12 or 0; for $a = 0$, $(b,c) \in \{0,3\} \times \{0,2\}$ gives $0, 12, -24, -12$; for $a = 3$, $(b,c) \in \{0,3\} \times \{1,3\}$ gives $12, 24, -12, 0$. So exactly six tuples survive: the four constant tuples, $(0,3,0,3,0)$ and $(3,0,3,0,3)$.

For each of these the lower differences also satisfy the conditions: constants are clear, and $(0,3,0,3,0)$ has differences $c_0, \ldots, c_4 = 0, 3, -6, 12, -24$, divisible by $0!, 1!, 2!, 3!, 4!$ respectively, so $3\binom{x}{1} - 6\binom{x}{2} +$

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$12\binom{x}{3} - 24\binom{x}{4} \in \mathbb{Z}[x]$; the last tuple comes from 3 minus this polynomial. The answer is 6. (Theorem 1 alone would have allowed 12.)

Answer: 6

Problem 3

Let p be an odd prime. Prove that

$$f(x) = x^{p-2} + 2x^{p-3} + 3x^{p-4} + \cdots + (p-2)x + (p-1)$$

is irreducible over $\mathbb{Q}$.

Write $f(x) = \sum_{k=1}^{p-1} k\, x^{p-1-k}$. Then

$$(x-1)f(x) = \sum_{k=1}^{p-1} k\, x^{p-k} - \sum_{k=1}^{p-1} k\, x^{p-1-k} = x^{p-1} + x^{p-2} + \cdots + x - (p-1),$$

since the coefficient of x^j for $1 \le j \le p-1$ is $(p-j) - (p-1-j) = 1$ and the constant term is $-(p-1)$. Multiplying by $x-1$ again,

$$(x-1)^2 f(x) = (x^p - x) - (p-1)(x-1) = x^p - px + (p-1).$$

Substituting $x+1$ for x:

$$x^2 f(x+1) = (x+1)^p - p(x+1) + p - 1 = \sum_{k=2}^{p} \binom{p}{k} x^k,$$

because the terms $1 + px$ from the binomial expansion cancel with $-px - p + p - 1$. Hence

$$g(x) = f(x+1) = x^{p-2} + \binom{p}{p-1} x^{p-3} + \cdots + \binom{p}{3} x + \binom{p}{2}.$$

For $2 \le k \le p-1$, p divides $\binom{p}{k} = \frac{p!}{k!(p-k)!}$, since p divides the numerator and not the denominator. The leading coefficient is 1, and the constant term $\binom{p}{2} = p \cdot \frac{p-1}{2}$ is not divisible by p^2 because $\frac{p-1}{2}$ is an integer less than p. By Eisenstein, g is irreducible over $\mathbb{Q}$. If $f = uv$ with u, v nonconstant, then $g(x) = u(x+1)v(x+1)$ would be a factorisation of g, so f is irreducible too. (For $p = 3$, $f = x + 2$ and the statement is trivial; the argument covers this case.)

Problem 4

Find all polynomials P with integer coefficients such that $P(n)$ divides n^n for every positive integer n.

The answer is $P(x) = \pm 1, \pm x$ and $\pm x^2$. These work: $n^k \mid n^n$ for $k \le 2$ when $n \ge 2$, and for $n = 1$ we need $P(1) \mid 1$, which holds since $P(1) = \pm 1$ for all six polynomials.

Conversely, suppose P works. Since $n^n \neq 0$, $P(n) \neq 0$ for all $n \ge 1$, so P is not the zero polynomial; write $P(x) = x^k Q(x)$ with $Q \in \mathbb{Z}[x]$ and $c = Q(0) \neq 0$. Then $Q(n) \mid P(n) \mid n^n$ for all $n \ge 1$.

Let q be a prime with $q > |c|$. Every prime factor of $Q(q)$ divides q^q, so $Q(q) = \pm q^j$ for some $j \ge 0$. By Theorem 1, $Q(q) \equiv Q(0) = c \pmod{q}$, and $q \nmid c$ because $0 < |c| < q$. Hence $j = 0$ and $Q(q) = \pm 1$. This holds for infinitely many primes q, so the polynomial $Q(x)^2 - 1$ has infinitely many roots and is zero; hence Q is the constant 1 or -1.

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So $P(x) = \pm x^k$, and the condition at $n = 2$ is $2^k \mid 4$, so $k \leq 2$. Together with the check above, the solutions are exactly ± 1, $\pm x$, $\pm x^2$.

Problem 5

Let P be a monic polynomial with integer coefficients, all of whose complex roots have absolute value 1. Prove that every root of P is a root of unity.

Let $n = \deg P$ and let $z_1, \ldots, z_n$ be the roots of P, with multiplicity, so $P(x) = \prod(x - z_i)$. For each positive integer k define $P_k(x) = \prod_{i=1}^{n}(x - z_i^k)$.

The coefficient of x^{n-j} in P_k is $(-1)^j e_j(z_1^k, \ldots, z_n^k)$, where e_j is the jth elementary symmetric function. As a function of $z_1, \ldots, z_n$ this is a symmetric polynomial with integer coefficients, so by the fundamental theorem of symmetric polynomials it is an integer polynomial in $e_1(z), \ldots, e_n(z)$, which are (up to sign) the coefficients of P and hence integers. So $P_k \in \mathbb{Z}[x]$ and is monic of degree n.

Since $|z_i^k| = 1$, $e_j(z_1^k, \ldots, z_n^k)$ is a sum of $\binom{n}{j}$ numbers of absolute value 1, so its absolute value is at most $\binom{n}{j}$. Hence each coefficient of P_k is an integer in a fixed bounded range, and there are only finitely many possible polynomials P_k as k varies.

Each P_k has at most n roots, and z_1^k is a root of P_k. So all the numbers $z_1, z_1^2, z_1^3, \ldots$ lie in the union of the root sets of finitely many polynomials, which is a finite set. Hence $z_1^k = z_1^l$ for some $k < l$, and since $z_1 \neq 0$, $z_1^{l-k} = 1$. The same argument applies to every root.

Problem 6

Let P be a nonconstant polynomial with integer coefficients and k a positive integer. Prove that there are k consecutive positive integers $n+1, n+2, \ldots, n+k$ such that each of $P(n+1), \ldots, P(n+k)$ is divisible by the square of some prime.

Let $R \in \mathbb{Z}[x]$ be a nonconstant factor of P that is irreducible over $\mathbb{Q}$ (by Theorem 4 we may take it in $\mathbb{Z}[x]$). Its derivative R' is nonzero of smaller degree, so a common factor of R and R' in $\mathbb{Q}[x]$ would be a factor of R of degree between 1 and $\deg R - 1$, which does not exist. Hence $\gcd(R, R') = 1$ in $\mathbb{Q}[x]$, and there are $U, V \in \mathbb{Q}[x]$ with $UR + VR' = 1$. Clearing denominators, $U_1 R + V_1 R' = M$ for some $U_1, V_1 \in \mathbb{Z}[x]$ and a positive integer M.

By Schur's theorem infinitely many primes divide some value $R(a)$ with a a positive integer. Choose k distinct such primes $p_1, \ldots, p_k$, none dividing M, and positive integers a_i with $p_i \mid R(a_i)$. Then $p_i \nmid R'(a_i)$, since otherwise $p_i \mid U_1(a_i)R(a_i) + V_1(a_i)R'(a_i) = M$.

For integers a and h we have $R(a+h) = R(a) + hR'(a) + h^2 S(a,h)$ with S an integer polynomial, because $(a+h)^m = a^m + ma^{m-1}h + h^2(\ldots)$. Since $p_i \nmid R'(a_i)$, choose an integer t_i with $\frac{R(a_i)}{p_i} + t_i R'(a_i) \equiv 0 \pmod{p_i}$ and put $b_i = a_i + t_i p_i$. Then $R(b_i) \equiv R(a_i) + t_i p_i R'(a_i) \equiv 0 \pmod{p_i^2}$, so $p_i^2 \mid R(b_i) \mid P(b_i)$.

The moduli $p_1^2, \ldots, p_k^2$ are pairwise coprime, so by CRT there is a positive integer n with $n + i \equiv b_i \pmod{p_i^2}$ for $i = 1, \ldots, k$. By Theorem 1, $P(n+i) \equiv P(b_i) \equiv 0 \pmod{p_i^2}$, so each $P(n+i)$ is divisible by the square of the prime p_i.